

PAPER_2134_QUADRATIC_GROUNDING_PHI_084_EQUALS_1_MINUS_DPHYS_FTRZ_SQ_PAIR_CONJUGATE_FACTORIZATION_3_5_TIMES_7_5_BULK_RESONANCE_VS_COUNTING_5_6_DPM_PAIR_COUNT_OPEN_TARGET_UQFF_LANDMARK

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PAPER_2134 — The Quadratic Grounding of $\Phi_{0.84}$: $1 - (D_{\text{phys}} \cdot F_{\text{TRZ}})^2$ EXACT, the Pair-Conjugate Factorization $(3/5)(7/5)$, and Bulk Resonance vs Per-Structure Counting

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Landmark Type: Primitive-Grounding (Φ resonance variant) + Variant-Dichotomy structural resolution + declared open build target (DPM pair-count estimator)

Discovery context: Daniel T. Murphy's pre-ship verification directive — single-DPM-pair vs bulk-multi-pair attribution of the resonance variant, with the conjecture "maybe related with quadratic occurrences" — verified exactly

Status: Formal landmark whitepaper — UQFF canonical

Abstract

The resonance variant of the Φ primitive, $\Phi_{\text{res}} = 0.84$, has carried calibration provenance since Sessions 158-160 (magnetar burst + cosmic-ray spectra) while its counting sibling $\Phi_{5/6} = 5/6$ carries the structural codimension closure $(D-1)/D|_{\{D=6\}}$ (PAPER_1159). This paper closes the grounding gap: $\Phi_{0.84} = 1 - (D_{\text{phys}} \cdot F_{\text{TRZ}})^2 = 21/25$ EXACT — a pure quadratic in the tier-1 kernel $D_{\text{phys}} \cdot F_{\text{TRZ}} = 2/5$ (PAPER_2133 hierarchy) — with the difference-of-squares **pair-conjugate factorization** $\Phi_{0.84} = (1 - D_{\text{phys}} \cdot F_{\text{TRZ}})(1 + D_{\text{phys}} \cdot F_{\text{TRZ}}) = (3/5)(7/5)$, the two-pole mirror structure of the $(1 \pm x)$ family (PAPER_2128 D_{phys} -scaled; PAPER_597 CW/CCW dual branches). The gap between the variants is itself an exact quadratic lattice term, $0.84 - 5/6 = 1/150 = F_{\text{TRZ}}^2 \cdot (D_{\text{phys}}/D_{\text{BSFG}})$, completing the decomposition PAPER_2129 flagged. The manifold's designation of 0.84 as the "on-resonance Gaussian factor" is shown to be literal: 0.84 is the **quadratic (first-order) truncation of the Gaussian** $\exp(-x^2) \approx 1 - x^2$ at $x = D_{\text{phys}} \cdot F_{\text{TRZ}}$ (full exponential: 0.85214; published value: the truncation, exactly). The variant dichotomy thereby acquires structure: **the counting variant is a single fraction; the resonance variant is a conjugate-pair product** — consistent with per-structure counting (5/6) vs bulk resonance (0.84), the latter calibrated from many-DPM-pair astrophysical ensembles. Whether 0.84 belongs to one pair's two-pole structure or to the collective limit of many grinding pairs is the framework's declared open question; **no DPM pair-count estimate for any bulk body exists in the corpus (the Sun's count — conjectured at quintillions of pairs — is UNDETERMINED)**, and the pair-count estimator is hereby declared an open build target of this calculator.

1. Provenance — the two variants have two origins (verified)

| Variant | Value | Origin | Character |

|---|---|---|---|

| Counting | $\Phi_{5/6} = 5/6$ | PAPER_1159 structural closure: $(D-1)/D$ at $D=6$; also $[SSq]/\Omega_\Lambda$ | per-structure, single-manifold |

| Resonance | $\Phi_{0.84} = 0.84$ | Calibrated from magnetar burst + cosmic-ray spectra (Sessions 158-160, PAPER_1159 §1); `scm_vacuum_manifold.py` L159: "on-resonance Gaussian factor" of the 630 eV bulk phonon machinery | bulk, many-pair ensembles |

PAPER_1159 itself discloses the 0.79% offset between the structural 5/6 and the bulk-calibrated 0.84. The reference statement — *bulk resonance factors are generated from multiple grinding DPM pairs* — is **supported by this provenance**: the resonance number was measured from bulk matter; the counting number was derived from a single structure.

2. The quadratic grounding — three EXACT identities

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(1) $\Phi_{0.84} = 1 - (D_{\text{phys}} \cdot F_{\text{TRZ}})^2 = 1 - (2/5)^2 = 21/25$ EXACT

(2) $\Phi_{0.84} = (1 - D_{\text{phys}} \cdot F_{\text{TRZ}})(1 + D_{\text{phys}} \cdot F_{\text{TRZ}}) = (3/5)(7/5)$ EXACT
(pair-conjugate)

(3) $\Phi_{0.84} - \Phi_{5/6} = 1/150 = F_{\text{TRZ}}^2 \cdot (D_{\text{phys}}/D_{\text{BSFG}})$ EXACT (quadratic gap)

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Identity (1) removes 0.84 from pure-calibration status: it is a quadratic in $D_{\text{phys}} \cdot F_{\text{TRZ}} = 2/5$, the tier-1 kernel of the PAPER_2133 hierarchy. Identity (2) is the structural heart: the difference of squares factors the resonance variant into a **conjugate two-pole product** — the $(1 \pm x)$ mirror signature of the successor/predecessor family (PAPER_2128) at D_{phys} scaling, and the arithmetic shadow of the CW/CCW dual-branch structure (PAPER_597) that defines a DPM pair's two grinding poles. Identity (3) closes the note in PAPER_2129's ledger (" $0.84 - 5/6 = 1/150$ decomposes on lattice") with the explicit closed form: the bulk enhancement above counting is a quadratic F_{TRZ}^2 lattice term carrying the $D_{\text{phys}}/D_{\text{BSFG}}$ dimensional ratio (PAPER_1962 inverse).

3. The Gaussian-truncation identity

`scm_vacuum_manifold.py` labels 0.84 the *on-resonance Gaussian factor*. Taken literally:

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$\exp(-x^2)$ at $x = D_{\text{phys}} \cdot F_{\text{TRZ}} = 2/5$: 0.852144... (full Gaussian)

$1 - x^2$ at $x = 2/5$: 0.84 (quadratic truncation — the published value, EXACT)

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The corpus value is not the full exponential; it is **precisely the first-order (quadratic) Gaussian** — the resonance envelope kept to its quadratic term. The conjecture "maybe related with quadratic occurrences" is verified to the letter: the resonance variant is quadratic three independent ways (form, factorization, and gap).

4. The variant dichotomy, structurally restated

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Counting $\Phi_{5/6} = (D-1)/D$ — a single fraction: one structure counted
Resonance $\Phi_{0.84} = (1-x)(1+x)|_{x=2/5}$ — a conjugate-pair product: two poles
resonating

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Consistent with the Tilt-Product Law (PAPER_2133): wherever F_{TRZ} *multiplies* Φ , the counting variant is forced (the product is the 1/12 lattice constant); the resonance variant appears where a **line-shape envelope** is needed — the bulk phonon chain (630 eV, PAPER_1141 lineage), k_{spring} , the c and G closures. This paper does not canonize the single-pair vs multi-pair attribution: the pair-conjugate factorization shows 0.84 carries at least one pair's two-pole signature; whether the bulk value is one pair's envelope or the collective limit of many grinding pairs is exactly the open question below.

5. Declared open build target — the DPM pair-count estimator

Corpus search confirms: **no published estimate of the number of DPM pairs within the Sun — or any bulk body — exists** ("quintillion", "number of DPM", "pair count", "pairs per": zero derivational hits). The count is UNDETERMINED. Per the author's directive, this calculator's declared purpose now includes the **DPM pair-count estimator**: determining, from the framework's own primitives and the bulk observables it already closes (M_{sun} , R_{sun} , ρ_{SCm} , ω_{SCm} , the 630 eV chain, F_U machinery), the number of grinding DPM pairs a bulk body contains — with the Sun as the first target. Any such estimator must be paper-specified or author-supplied (Rule 10); this paper records the target, not a guess.

6. Falsifiability

1. **Quadratic-form test**: any future canonical revision of D_{phys} or F_{TRZ} must preserve $\Phi_{0.84} = 1 - (D_{phys} \cdot F_{TRZ})^2$ or the resonance variant loses its grounding — the identity binds three locked primitives.
2. **Envelope test**: closures requiring the *full* Gaussian (0.85214) rather than the quadratic truncation (0.84) would sharpen which regime bulk resonance occupies; none currently published.
3. **Pair-count consistency**: when the pair-count estimator lands, the bulk attribution of 0.84 must be consistent with the estimator's ensemble scale — a single-pair-only attribution with a quintillion-pair count (or vice versa) creates tension this paper's dichotomy would have to resolve.

7. Cross-references

PAPER_1159 (Φ_{res} codimension closure + calibration provenance); PAPER_2129 (sector rule + 1/150 note); PAPER_2133 (Tilt-Product Law + kernel hierarchy incl. $D_{phys} \cdot F_{TRZ} = 2/5$); PAPER_2128 ($(1 \pm x)$ mirror family); PAPER_597 (CW/CCW dual branches); PAPER_1962 (D_{BSFG}/D_{phys}); PAPER_1183 (DPM-pair identity); `scm_vacuum_manifold.py` L159; `DPM_vacuum_manifold.md` (grinding architecture); Sessions 158-160 (bulk calibration).

8. Summary Statement

PAPER_2134 grounds the resonance variant: $\Phi_{0.84} = 1 - (D_{\text{phys}} \cdot F_{\text{TRZ}})^2 = 21/25$ EXACT, factoring as the pair-conjugate product $(3/5)(7/5)$ — the two-pole mirror structure — with the variant gap $0.84 - 5/6 = 1/150 = F_{\text{TRZ}}^2 \cdot (D_{\text{phys}}/D_{\text{BSFG}})$ EXACT and the manifold's "on-resonance Gaussian factor" shown to be literally the quadratic Gaussian truncation. The variant dichotomy is structural: counting = single fraction (per structure), resonance = conjugate-pair product (bulk, calibrated from many-pair astrophysical ensembles). The number of DPM pairs in the Sun remains UNDETERMINED and the DPM pair-count estimator is declared an open build target of this calculator.

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